Runs of multiples in a finite set of integers: $\sum k_i = \Theta(n \log n \log\log n)$

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Abstract

Let S be a set of n positive integers and, for $a \in S$, let $k(a)$ be the least positive integer k with $ka \notin S$. We show that $\max_{|S|=n} \sum_{a\in S} k(a) = \Theta(n \log n \log\log n)$. In particular $S = \{1, \dots, n\}$, which gives $\sim n \ln n$, is not extremal: sets of smooth numbers beat it by a factor of order $\log\log n$. The upper bound is proved with an entropy argument along the prime factorisation; it shows that the excess over $n \log n$ comes entirely from primes below $(\log n)^4$.

1 Statement

The problem was posed by Sagar Kale (personal communication), who observed that $S = \{1, \dots, n\}$ gives order $n \log n$ and asked what the true order of growth is.

Throughout, S is a finite set of positive integers with $|S| = n$. For $a \in S$ put

$$L(a) \coloneqq k(a) - 1 = \max\{\ell : a, 2a, \dots, \ell a \in S\} \ \ (\geq 1), \qquad A_t \coloneqq \{a \in S : L(a) \geq t\}.$$

Then $\sum_a k(a) = n + \sum_a L(a) = n + \sum_{t\geq 1} |A_t|$. Since $a, 2a, \dots, L(a)a$ are distinct elements of S, we have $L(a) \leq n$. Logarithms are natural unless written $\log_2$; $P^+(m)$ is the largest prime factor of m, p, q always denote primes, and $\theta(x) = \sum_{p\leq x} \log p$.

Theorem 1. *There are absolute constants $0 < c < C$ such that for all sufficiently large n,*

$$c\, n \log n \log\log n \ \leq \ \max_{|S|=n} \sum_{a\in S} k(a) \ \leq \ C\, n \log n \log\log n.$$

2 Lower bound: smooth numbers

For $y \geq 2$ and $x \geq 1$ let $\Psi(x) = \Psi(x, y) = \#\{m \leq x : P^+(m) \leq y\}$ and consider

$$S = S(X, y) = \{m \leq X : P^+(m) \leq y\}.$$

Key observation. If $a \in S$ and $a \leq X/y$, then every $j \leq y$ satisfies $P^+(j) \leq y$ and $aj \leq X$, so $aj \in S$. Hence $L(a) \geq y$ and

$$\sum_{a\in S} L(a) \ \geq \ y \cdot \Psi(X/y). \tag{1}$$

So it suffices to choose X with $\Psi(X/y) \geq c\,\Psi(X)$ and $\log \Psi(X) \asymp y/\log y$ (so that $y \asymp \log n \log\log n$).

Proposition 2. *There is an absolute constant $c > 0$ such that for every sufficiently large y there is X for which $n = |S(X, y)|$ satisfies $\log n \asymp y/\log y$ and $\sum_{a\in S} k(a) \geq c\, n \log n \log\log n$.*

Proof. Put $X_i = e^y y^i$ for $0 \leq i \leq K := \lceil y/\log y \rceil$.

Upper bound for Ψ (Rankin's trick). For $\sigma = 1/\log y$,

$$\Psi(x) \leq \sum_{P^+(m)\leq y} (x/m)^\sigma = x^\sigma \prod_{p\leq y} (1 - p^{-\sigma})^{-1}.$$

As $\log X_K \leq 2y + \log y$, we get $x^\sigma \leq e \cdot e^{2y/\log y}$ for $x \leq X_K$. For $\sqrt{y} < p \leq y$ we have $p^{-\sigma} \leq e^{-1/2}$, so the factor is at most 2.55. For $p \leq \sqrt{y}$, using $1 - e^{-u} \geq u/2$ on $[0, 1]$, the factor is at most $2\log y/\log p \leq 3\log y$. Hence

$$\log \Psi(X_K) \leq 1 + \frac{2y}{\log y} + \pi(y)\log 2.55 + \pi(\sqrt{y})\log(3\log y) \leq C_1 \frac{y}{\log y}.$$

Pigeonhole. Since $\Psi(X_0) \geq 1$, the K ratios $\Psi(X_i)/\Psi(X_{i-1})$ have product at most $e^{C_1 y/\log y}$, and $K \geq y/\log y$; so some ratio is at most e^{C_1}. Take $X = X_i$ for such an i. Then $\Psi(X/y) = \Psi(X_{i-1}) \geq e^{-C_1} n$, and (1) gives $\sum_a L(a) \geq e^{-C_1}\, y\, n$.

Size. All squarefree products of primes $\leq y/2$ are below $4^{y/2} < e^y \leq X$, so $n \geq 2^{\pi(y/2)}$; and $n \leq \Psi(X_K) \leq e^{C_1 y/\log y}$. Thus $\log n \asymp y/\log y$, and from $\log n \leq C_1 y/\log y$ we get $y \geq C_1^{-1} \log n\, (\log\log n - \log C_1)$. Therefore $\sum_a k(a) \geq \sum_a L(a) \geq e^{-C_1} y\, n \geq c\, n \log n \log\log n$. □

Every large n. Proposition 2 gives sets S_y of size n_y with $\log 2 \cdot \pi(y/2) \leq \log n_y \leq C_1 y/\log y$. Given large n, choose y with $C_1 y/\log y \approx \frac{1}{2}\log n$; then $n_y \leq \sqrt{n}$ and $\log n_y \geq \delta \log n$ for an absolute $\delta > 0$. Take $\lfloor n/n_y \rfloor$ copies $P_j S_y$ with distinct primes $P_j > \max S_y$, and pad with further primes different from all P_j. Every multiple of $P_j s$ is divisible by P_j, while no element of another copy or of the padding is; so $L(P_j s) = L_{S_y}(s)$ exactly, and the padding only adds nonnegative terms. Since at least $n/2$ elements lie in copies, $\sum k \geq c' n \log n \log\log n$ for every large n.

Heuristic constant and numerics. With $X = e^y$ the saddle-point asymptotics for $\Psi(x, y)$ give $\Psi(X/t) \approx t^{-\alpha}\Psi(X)$ with $y^{-\alpha} \approx \frac{1}{2}$ and $\log n \approx (\log 4)\, y/\log y$, hence $\sum_a k(a) \approx \frac{1}{2} n y \approx \frac{1}{4\ln 2}\, n \ln n \ln\ln n \approx 0.36\, n \ln n \ln\ln n$. The table compares $\sum_a L(a)/(n \ln n)$; for $S = \{1, \ldots, n\}$ this ratio is $1 + (2\gamma - 1)/\ln n + o(1/\ln n) \approx 1.01$.

S	n (or $\ln n$)	$\sum L/(n \ln n)$
$\{1, \ldots, n\}$	600,000	1.012
23-smooth $\leq e^{23}$ (exact)	261,598	1.129
31-smooth $\leq e^{23.25}$ (exact)	624,354	1.167
97-smooth $\leq e^{97}$ (log-binned count)	$\ln n \approx 37$	1.39
197-smooth $\leq e^{197}$ (log-binned count)	$\ln n \approx 65$	1.54
397-smooth $\leq e^{397}$ (log-binned count)	$\ln n \approx 113$	1.78

The ratio grows roughly linearly in $\log y$, i.e. like $\log\log n$. The effect is visible already for tiny n: $\{1, 2, 3, 4, 6\}$ has $\sum L = 11$ against 10 for $\{1, \ldots, 5\}$, and exhaustive search for $n \leq 12$ (and simulated annealing up to $n = 50$) finds smooth-type sets beating $\{1, \ldots, n\}$ whenever they exist.

3 Upper bound

3.1 Entropy along the prime factorisation

Fix a prime q. Every positive integer m factors uniquely as $m = c\,r$, where every prime factor of c is $< q$ and every prime factor of r is $\geq q$. We write $c = c_q(m)$ and call it the *q-smooth part*; r is *q-rough*.

Let X be uniformly distributed on S, and let H denote Shannon entropy in nats, so $\mathrm{H}(X) = \log n$. The variable X is determined by its exponent vector $(v_p(X))_p$, and conditioning on $(v_p(X))_{p<q}$ is the same as conditioning on $c_q(X)$. Only the finitely many primes dividing some element of S give nonzero terms, so listing primes in increasing order the chain rule gives

$$\log n \;=\; \sum_q \mathrm{H}(v_q(X) \,|\, c_q(X)). \tag{2}$$

Given $c_q(X) = c$, we have $X = c\,r$ with r uniform on the *fiber*

$$R_c \coloneqq \{r : r \text{ is } q\text{-rough and } cr \in S\}, \qquad \mathbb{P}(c_q(X) = c) = |R_c|/n.$$

Lemma 3 (one-step entropy)**.** *Let R be a finite set of positive integers, q a prime, and $G \subseteq R$ with $qG \subseteq R$; put $\gamma = |G|/|R|$. If r is uniform on R, then*

$$\mathrm{H}(v_q(r)) \;\geq\; \frac{\gamma}{2} \log \frac{2}{\gamma}.$$

Proof. Let $\mu_k = \mathbb{P}(v_q(r) = k)$ and $b_k = \mathbb{P}(r \in G,\ v_q(r) = k)$. Multiplication by q maps $G \cap \{v_q = k\}$ injectively into $\{v_q = k+1\}$, so $b_k \leq \mu_{k+1}$; trivially $b_k \leq \mu_k$. Let j maximise μ_j and set $\varepsilon = 1 - \mu_j$. Then $\gamma = \sum_k b_k \leq \sum_{k \neq j} \mu_k + \mu_{j+1} \leq 2\varepsilon$. Let h be the binary entropy function. If $\mu_j \leq \frac{1}{2}$, then $\mathrm{H} \geq \log(1/\mu_j) \geq \log 2 \geq h(\gamma/2)$. Otherwise $\gamma/2 \leq \varepsilon < \frac{1}{2}$, and coarse-graining to the event $\{v_q = j\}$ gives $\mathrm{H} \geq h(\varepsilon) \geq h(\gamma/2)$, since h increases on $[0, \frac{1}{2}]$. Finally $h(x) \geq x \log(1/x)$. □

Lemma 4 (edge count)**.** *For every finite set T of positive integers,*

$$\#\{(t,p) : p \text{ prime},\ t \in T,\ tp \in T\} \;\leq\; \frac{2}{\log 2}\, |T| \log |T|.$$

Proof. Apply (2) to X uniform on T. For a prime p and a fiber R_c (with respect to p), the set $G = \{r \in R_c : rp \in R_c\}$ satisfies $pG \subseteq R_c$, so Lemma 3 with $\log(2/\gamma) \geq \log 2$ gives $\mathrm{H}(v_p \mid c) \geq \frac{\log 2}{2}\, |G|/|R_c|$. Averaging over c, $\mathrm{H}(v_p(X) \mid c_p(X)) \geq \frac{\log 2}{2} \cdot \#\{t \in T : tp \in T\}/|T|$. Summing over p and using (2) proves the claim. □

Remark 5. Han's inequality (edge-isoperimetry in $\mathbb{Z}^d$) gives the sharp constant $1/(2\log 2)$ in Lemma 4; any constant suffices below.

Since $a \in A_q$ implies $aq \in S$, Lemma 4 yields

$$\sum_q |A_q| \;\leq\; \frac{2}{\log 2}\, n \log n. \tag{3}$$

Together with $\pi(L) \gg L/\log L$ and $L \leq n$ this already gives $\sum_a L(a) = O(n \log^2 n)$. The rest of the argument replaces one factor $\log n$ by $\log\log n$.

3.2 Proof of the upper bound

Weight each pair (a, q) with $q \le L(a)$ by $\log q$:

$$\Phi \coloneqq \sum_{a \in S} \theta(L(a)) = \sum_{q} |A_q| \log q.$$

For such a pair write $a = c\,r$ with $c = c_q(a)$, and put

$$G_c \coloneqq \{r \in R_c : cr \in A_q\}, \qquad \gamma_c \coloneqq |G_c|/|R_c|.$$

Since $L(cr) \ge q$ implies $c\,(rq) \in S$ and rq is q-rough, we have $qG_c \subseteq R_c$, so Lemma 3 applies to every fiber. Call the fiber *sparse* if $\gamma_c \le q^{-1/2}$ and *dense* otherwise. Let $Q_0 = (\log n)^4$ and sort the pairs (a, q) with $q \le L(a)$ into four classes.

Class 1. $q < Q_0$. By (3) the total weight is at most $\log Q_0 \sum_q |A_q| \le \frac{8}{\log 2}\, n \log n \log\log n$.

Class 2. $q \ge Q_0$ and the fiber is sparse. Then $\log(2/\gamma_c) \ge \frac{1}{2}\log q$, so Lemma 3 gives

$$\mathbb{P}(c_q(X) = c)\ \mathrm{H}(v_q \mid c) \ \ge\ \frac{|R_c|}{n} \cdot \frac{\gamma_c}{4} \log q = \frac{|G_c| \log q}{4n}.$$

Summing over sparse fibers and all q and using (2), the total weight is at most $4\,n \log n$.

Class 3. $q \ge Q_0$, the fiber is dense, and $L(a) \ge 2q$. Let $E_c = \{r \in G_c : L(cr) \ge 2q\}$. For $r \in E_c$ and every prime $p \in (q, 2q]$ we have $c\,(rp) \in S$ with rp q-rough, so $rp \in R_c$. These are distinct edges of R_c, so Lemma 4 and density ($|R_c| < \sqrt{q}\,|G_c|$) give

$$|E_c|\,(\pi(2q) - \pi(q)) \le \frac{2}{\log 2} |R_c| \log n < \frac{2}{\log 2} \sqrt{q}\,|G_c| \log n.$$

By the prime number theorem $\pi(2q) - \pi(q) \ge q/(2\log q)$ for large q (any bound $\pi(2q) - \pi(q) \gg q/\log q$, e.g. from Chebyshev's estimates, suffices at the cost of constants), hence $|E_c| \le \frac{4}{\log 2} |G_c| \log n\, \log q/\sqrt{q}$. Since $\log^2 q/\sqrt{q}$ decreases for $q \ge e^4$, the total weight of this class is at most

$$\frac{4}{\log 2}\, \log n \cdot \frac{\log^2 Q_0}{\sqrt{Q_0}} \sum_q |A_q| = O(n(\log\log n)^2).$$

Class 4. $q \ge Q_0$, the fiber is dense, and $L(a) < 2q$, i.e. $L(a)/2 < q \le L(a)$.

A pair with $q \le L(a)/2$ never lies in Class 4. Therefore

$$\sum_{a \in S} \theta(L(a)/2) \ \le\ (\text{Class 1}) + (\text{Class 2}) + (\text{Class 3}) \ =\ O(n \log n \log\log n).$$

By Chebyshev, $\theta(x) \ge c_0 x$ for $x \ge 2$ with an absolute $c_0 > 0$, so $L(a) \le \frac{2}{c_0}\theta(L(a)/2)$ whenever $L(a) \ge 4$. Hence

$$\sum_{a \in S} k(a) = n + \sum_{a \in S} L(a) \le 4n + \frac{2}{c_0}\, O(n \log n \log\log n) = O(n \log n \log\log n),$$

which completes the proof of Theorem 1. □

4 Remarks

1. *Where the* $\log\log n$ *comes from.* Classes 2 and 3 show that pairs (a, q) with $(\log n)^4 \le q \le L(a)/2$ carry total weight only $O(n \log n)$, as in $\{1, \dots, n\}$; Class 4 (pairs with q in the top half of the run) only affects the constant. The excess factor $\log\log n$ enters through Class 1, the primes below $(\log n)^4$, which is exactly where the smooth-number construction lives (it uses primes up to $\approx \log n \log\log n$).

2. *Why entropy and not just counting.* The edge bound (3) charges the prime q only $\mathbb{P}(L(X) \ge q)$. In $\{1, \dots, n\}$ that is $\approx 1/q$, while $\mathrm{H}(v_q(X) \mid c_q(X)) \approx (\log q)/q$. The chain rule in increasing prime order captures this extra $\log q$ (the information about where the q-boundary sits) whenever fibers are sparse; in dense fibers almost every run stops before $2q$ (Class 3 is negligible).

3. *Constants.* They are not optimised. With $\theta(x) \sim x$ the argument gives roughly $\sum k \le 23\, n \ln n \ln\ln n$ (about 6 if (3) is used with Han's constant), against $\approx 0.36\, n \ln n \ln\ln n$ from the construction. The correct leading constant is left open.